

MACAD 2025/26 · DIGITAL TOOLS FOR GRAPH MACHINE LEARNING

GraphML_Go2

Graph Learning Pipeline for Architectural Floor Plans

FINAL ASSIGNMENT

IAAC · MaCAD 2025/26

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Contents

Dataset exploration, research context, graph structure, building selection, and first results from graph analysis on the Narkomfin Building.

01 Dataset

Modified Swiss Dwellings: 5,372 residential floor plans as graphs

02 Research

MSD benchmark paper (ECCV 2024) · GNN node classification paper (eCAADe 2024)

03 Graph Structure

graph_in vs graph_out: zone labels, connectivity, and room type prediction

04 Pipeline

Floor plan design → graph construction & analysis → node classification

05 The Narkomfin Building

Building introduction · spatial concept · why it maps to graph ML

01 DATASET

Modified Swiss Dwellings

- 5,372 floor plans
- Multi-apartment clusters
- Corridor buildings
- Single units
- Graph modality only



3 sample plans, wide variety of residential typologies, scales and orientations

5,372

Floor Plans

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Zone Types

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Room Types

.pickle

Graph Format

02 RESEARCH

Literature

PAPER 1

MSD: A Benchmark Dataset for Floor Plan Generation of Building Complexes

ECCV 2024 · TU Delft

- 5,372 Swiss residential floor plans
- 3 modalities: image, geometry, graph
- Richer than RPLAN & LIFULL: multi-apartment, compass orientation, inter-unit links
- Graph encodes spatial zones, not room types

This dataset is our primary data source.

PAPER 2

GNN for Node Classification and Attribute Allocation in Architectural BIM

eCAADe 2024 · Cardiff University

- Uses the same MSD dataset
- Converts floor plans to graphs via TopologicPy
- Benchmarks GCN, GAT, GraphSAGE architectures
- GraphSAGE-Pool, 4 layers, ~95% accuracy

The pretrained model from this paper is what we will run on our floor plan.

03

GRAPH STRUCTURE

Data Structure

graph_in: model input

Attribute	Type	Values
zoning_type	int	0 · 1 · 2 · 3
connectivity	str	'door' · 'entrance'

graph_out: ground truth

Attribute	Type	Description
room_type	int	0 – 8
geometry	Polygon	2D room outline
centroid	Point	centre of room

Room types (0 – 8)

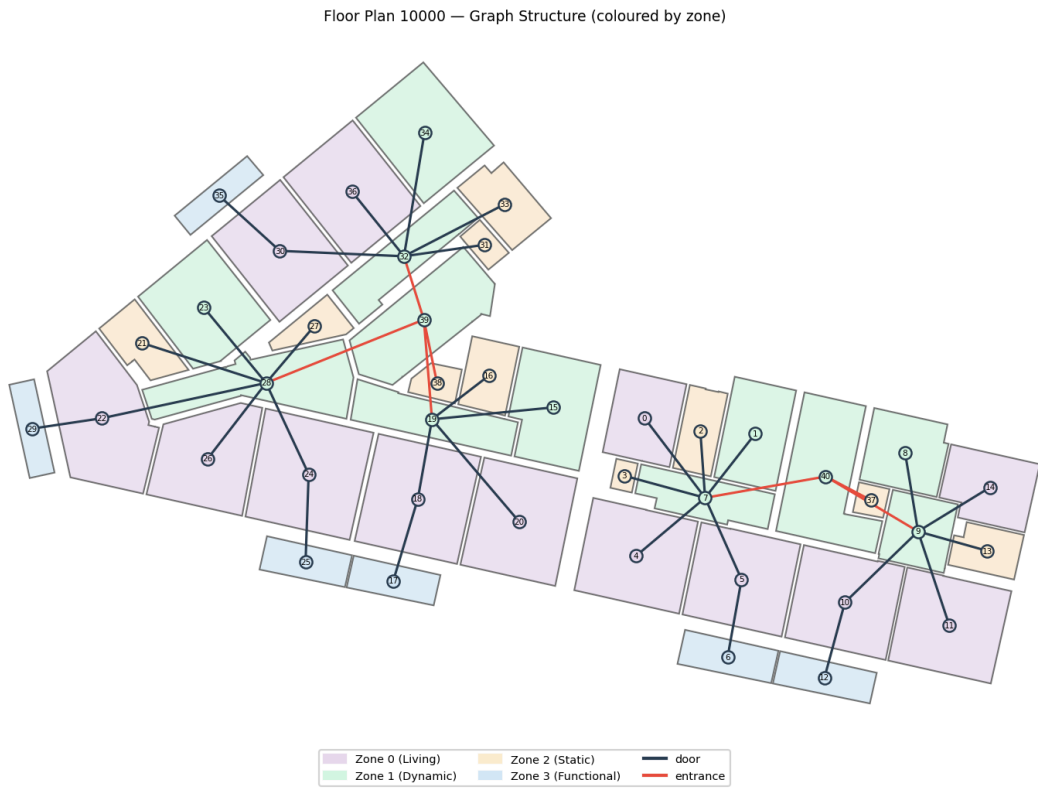
ID	Room	ID	Room
0	Balcony	5	Kitchen
1	Bathroom	6	Living Room
2	Bedroom	7	Storeroom
3	Corridor	8	Other
4	Dining		

Zone types (0 – 3)

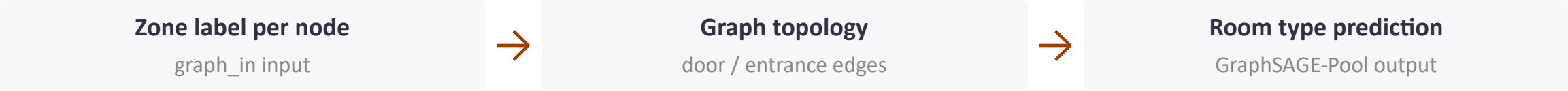
ID	Zone	Typical rooms
0	Living	Living room, dining
1	Dynamic	Corridor, entrance
2	Static	Bedroom, bathroom
3	Functional	Kitchen, storeroom

03 GRAPH STRUCTURE

From Plan to Graph: Plan 10000



Plan 10000: room types (left) and spatial connectivity graph (right)



04 NEXT STEPS

Pipeline

Floor Plan Input

- 01** Use the F-type and K-type floor plans from the Narkomfin Building as our input geometry, redrawn in Rhino as closed polylines with shared edges for TopologicPy adjacency detection.

Graph Construction & Analysis / TopologicPy

- 02** Convert geometry to a graph, label nodes with `zoning_type` and edges as door/entrance, then compute degree, betweenness, and closeness centrality.

Node Classification / GraphSAGE-Pool

- 03** Run the pretrained model to predict `room_type` from zone labels and connectivity. Compare results against MSD ground truth.

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OUR BUILDING

The Narkomfin Building



Post-restoration render · Ginsburg Architects · Moscow, 2021

ARCHITECTS	YEAR / LOCATION	TYOLOGY	SCALE	UNIT TYPES
Moisei Ginzburg & Ignaty Milinis	1930 · Moscow, Russia	Collective housing, dom- kommuna	54 apartments · 6 floors	F-type duplex · K- type communal

A landmark of Soviet Constructivism, designed as a **social condenser**: private space minimised, shared infrastructure maximised.

Two distinct floor plan typologies and a clear spatial hierarchy make it an ideal subject for graph-based analysis.

05 OUR BUILDING

Why Narkomfin

SPATIAL CONCEPT

The Social Condenser

- Private cells minimal: sleep and work only
- Shared corridor spine: the sole circulation artery
- Communal programme in a separate linked block
- Legible spatial hierarchy, ideal for graph analysis

WHY IT FITS GRAPH ML

Clear Zoning · Two Plan Types

- Corridor → **Dynamic (1)**, unambiguous node label
- Cells → **Static (2)**, communal → **Living (0) + Functional (3)**
- F-type & K-type units: two spatial configurations to compare
- Orthogonal geometry → clean TopologicPy graph construction

Soviet Constructivism · 1930

Social condenser

Linear spine

F-type duplex

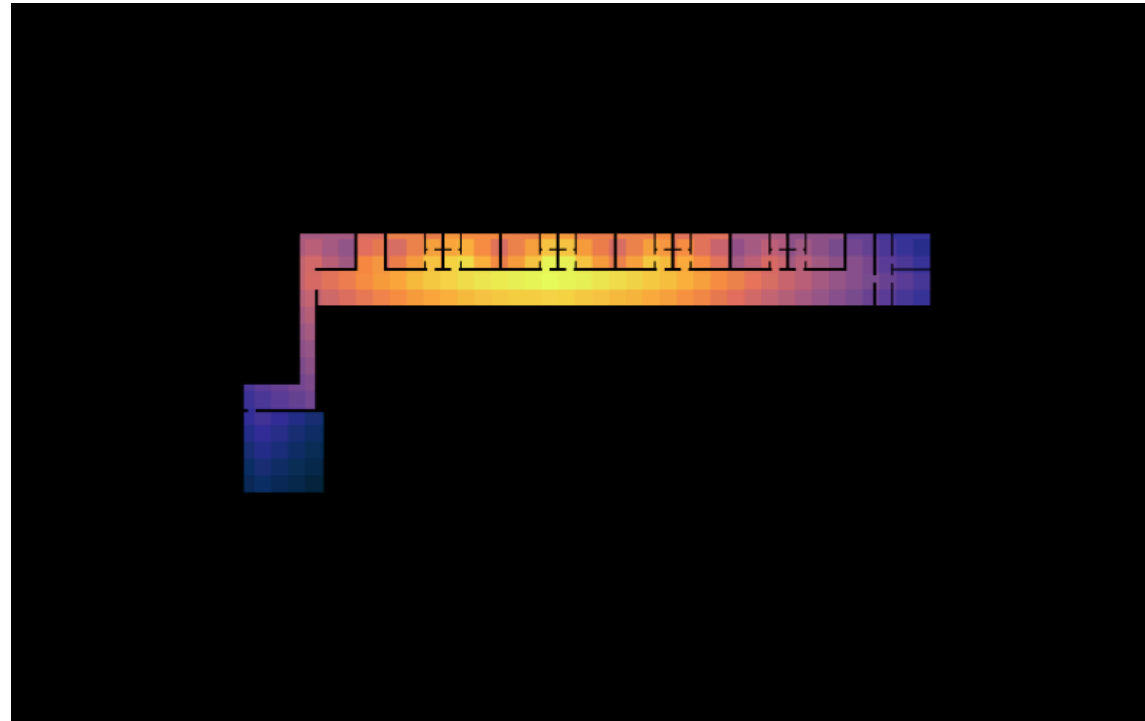
K-type communal

Modular repetition

06

GRAPH ANALYSIS

Plan 1 — Closeness Centrality



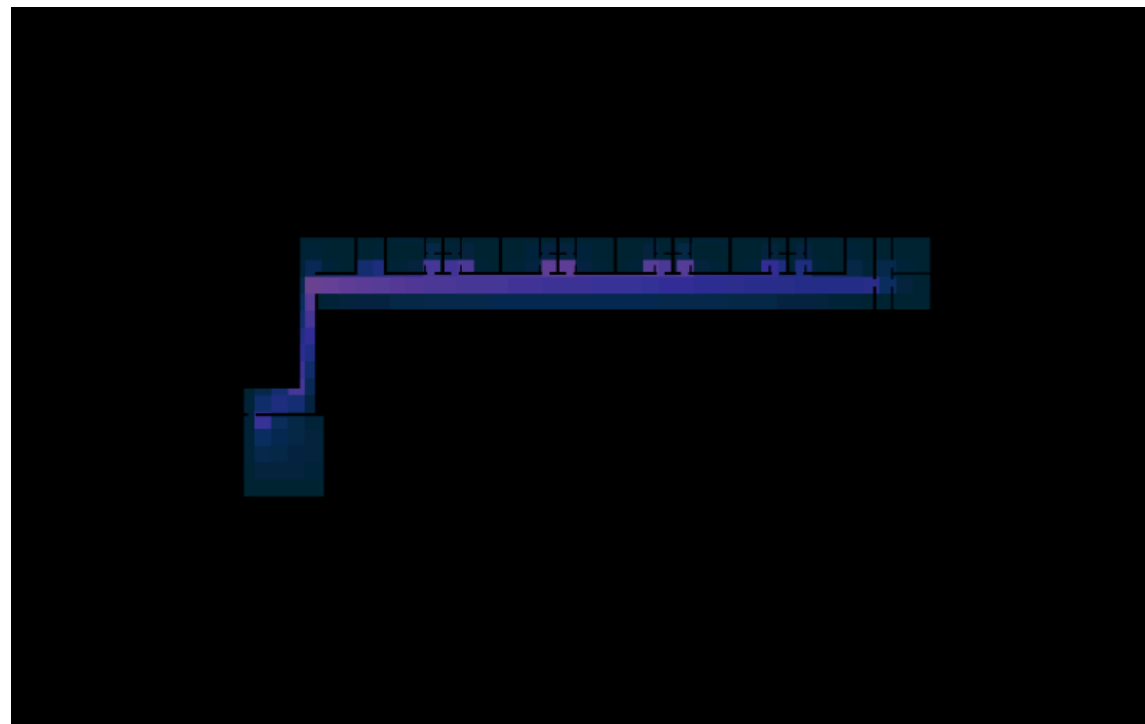
Thermal scale — warm = high integration, cool = low integration

The **corridor scores highest** — every room is reachable from it in the **fewest steps**. Closeness drops into the **private cells**, and the staircase block scores **lowest** as a dead-end terminal. The gradient confirms the corridor as the **social and circulatory core** of the plan.

06

GRAPH ANALYSIS

Plan 1 — Betweenness Centrality



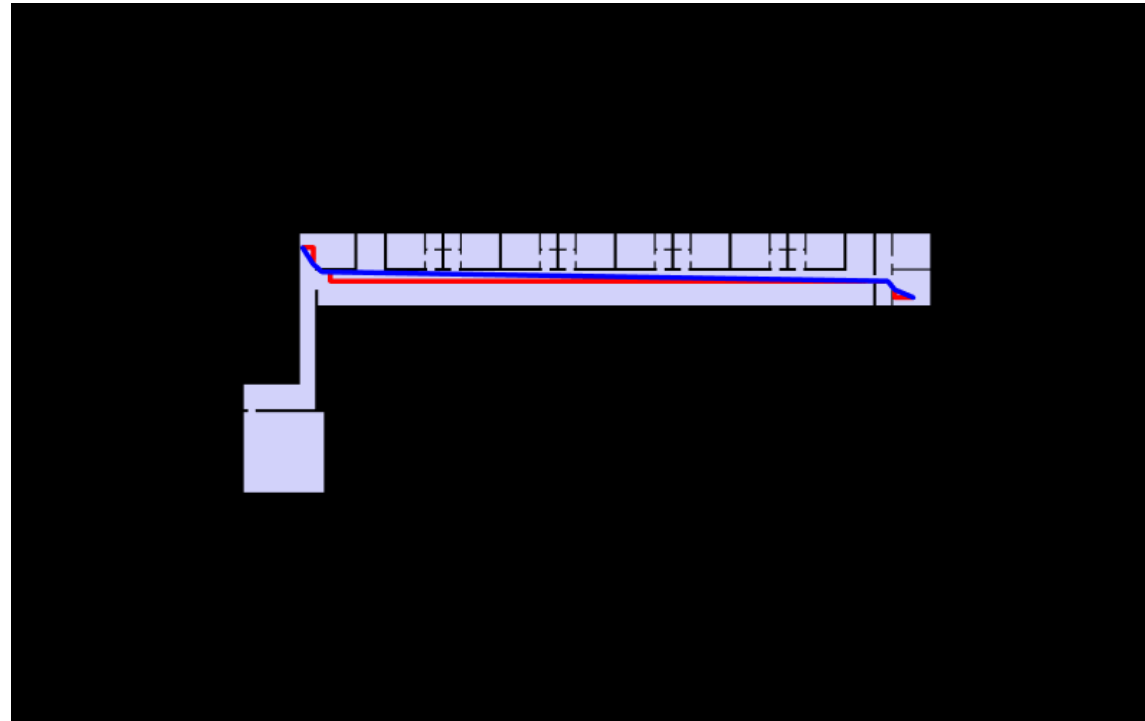
Thermal scale — warm = high traffic, cool = low traffic

Central corridor sections score highest — every cross-building journey passes through them. End rooms have **near-zero betweenness**; they are **destinations, not conduits**. The corridor serves as both the **most accessible and most traversed spine** of the plan.

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GRAPH ANALYSIS

Plan 1 — Shortest Path



Grid path (red, 77.2 units) vs straightened path (blue, 74.4 units)

The grid path (red, 77.2) and straightened path (blue, 74.4) are **nearly identical** — only **3.7% shorter** when straightened. The corridor is already the straight line, leaving **little room for a shortcut**. The **most efficient route** and the **only social route** are the same.